Vinayaka A Belludi

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Education

Kle Technological University **Jan. 2021 – Present**
Bachelor of Engineering in Electronics and Communication: CGPA - 7.4 *Hubli, Karnataka*

Sir MV Pu Collage **2018 – 2020**
Pre-University in PCMC : Score - 91 *Davangere, Karnataka*

Sri Taralabalu Central School **2018**
Score - 80 *Davangere, Karnataka*

Technical Skills

Languages: Python, C/C++
Frameworks:Pytorch, OpenCV, Pandas, Numpy, O3D, Scikit-Learn
Platforms: Linux, Visual Studio Code, Jupyter Notebook, Unity
Skills: AI/ML-DL, Computer Vision, 3D Reconstruction, Image Processing, Data Science, Game Development

Experience

Arka Creatos **June 2023 – November 2023**
Computer Vision Intern *Hubli, Karnataka*

- Developed an Oral Cancer Segmentation Model, and achieved a mean Intersection over Union (mIoU) score of 0.47.
- Implemented automated reporting systems for doctors, reducing report generation time by 70 percent
- Designed and developed a user-friendly application interface using PyQt5 for doctors, enabling real-time photo and video capture with integrated cancer segmentation capabilities and automated report generation

Projects

Neural Implicit 3D Reconstruction for Indoor Scenes | *PyTorch, NeRFs, 3D Reconstruction* **June 2024**
∗ Implemented the paper MonoSDF which utilizes NeRFs to achieve high-quality 3D reconstructions from 2D images
∗ Employed depth and normal maps to refine the spatial understanding of 3D scenes, enhancing the accuracy by 15 percent
∗ Generated SDFs from volumetrically rendered ray distances for 3D object boundaries
∗ Currently working on novelty to enhance further the impact and accuracy of the 3D reconstruction project.

Denoising 3D Point Cloud using Implicit Functions | *Pytorch, Point Clouds, Noise Removal* **February 2024**
∗ Implemented the paper Points2Surf which denoises 3D Point Cloud using Implicit Functions
∗ Used Signed Distance Function (SDF) estimation to remove noise and refine surface representations, significantly improving the clarity of the 3D point clouds
∗ Applied the Marching Cube algorithm for efficient point sampling for accurate representation of 3D surfaces

Semantic Segmentation on IDD | *Tensorflow, Semantic Segmentation, Image Processing,* **May 2023**
∗ Employed deep learning models for semantic segmentation of the Indian driving dataset.
∗ Trained a UNet model for 150 epochs, achieving a training accuracy of 38 percent. Trained a PSPNet model for 50 epochs, achieving a significantly higher training accuracy of 78 percent.

Research Publications

"AfforDrive: Detection of Drivable Area for Autonomous Vehicles" **December 2023**
∗ Presented at PReMI 2023, available in Springer. This paper introduces AfforDrive, a novel approach using Visual Affordance for accurately detecting drivable areas crucial for autonomous vehicles, contributing advancements in autonomous navigation and safety.

"VisionCraft: Advanced Image Captioning with Pre-trained Models" **June 2024**
∗ Accepted at ICMISC 2024, to be published in Springer. This paper presents a multi-modal approach to image captioning using pre-trained models, leveraging their data representations for improved caption generation.

Current Research **Ongoing**
∗ Currently researching 3D reconstruction using Neural Radiance Fields (NeRFs) multi-modal data integration for enhanced representation.
∗ Multi-modal data integration of image and text for Image Converser

Certifications & Achievements

Game Development Mini Degree **July, 2022**
∗ XAMK Open University, Finland. Earned 25 ECTS credits and successfully developed 3 games.

CVPR Workshop Participation **June 2023**
∗ Contributed to a semantic segmentation project with 27 classes and a dataset of 30,000 images.